GG-C-131C
April 18, 1977
SUPERSEDING
Int. Fed. Spec.
GG-C-00131B(DSA-DM)
November 24, 1972 and
Fed. Spec. GG-C-131
April 13, 1951

FEDERAL SPECIFICATION

CASSETTES, RADIOGRAPHIC FILM (MEDICAL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers non-phototiming cassettes; conventional and grid front types, having high speed intensifying screens (see 6.1).

1.2 Classification.

1.2.1 Types. The cassettes shall be of the following types, as specified (see 6.2):

Type I - Conventional; with high speed intensifying screens.

Type II - Grid front; with high speed intensifying screens.

1.2.2 Sizes. The cassettes shall be of the following sizes:

Size 1 - 8 by 10 inches
Size 2 - 10 by 12 inches
Size 3 - 11 by 14 inches
Size 4 - 14 by 17 inches

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 6525

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Federal Specifications:

L-P-509	- Plastic Sheet, Rod, and Tube, Laminated, Thermosetting.
GG-S-176	- Screen, X-Ray, Intensifying, Roentgenographic, for Diagnostic Radiography.
QQ-L-201	- Lead Sheet.
QQ-M-44	- Magnesium Alloy Plate and Sheet.
QQ-A-200/8	- Aluminum Alloy, Bar, Rod, Shapes, and Tube, Extruded, 6061 and 6062.
QQ-A-225/1	- Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn, or Cold Finished, 1100.
QQ-A-225/2	- Aluminum Alloy Bar, Rod and Wire, Rolled, Drawn, or Cold Finished, 3003.
QQ-A-250/1	- Aluminum Alloy 1100, Plate and Sheet.
QQ-A-250/2	- Aluminum Alloy 3003, Plate and Sheet.
QQ-A-250/8	- Aluminum Alloy 5052, Plate and Sheet.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.
PPP-B-566	- Boxes, Folding, Paperboard.

Federal Standards:

Fed. Std. No. 66	- Steel, Chemical Composition and Hardenability.
Fed. Std. No. 123	- Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 151	- Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Specification:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vapor Proof or Waterproof, Flexible.

MIL-P-26514 - Polyurethane Foam, Rigid or Plastic, for Packaging

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

PH3.21 Medical X-Ray Film Cassettes (Inch and Centimeter sizes), Dimensions for.

(Copies may be obtained from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish two sample units for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Material.

3.2.1 Corrosion-resisting steel. Corrosion-resisting steel shall conform to the requirements of type 300 series of Fed. Std. No. 66.

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3.2.2 Aluminum alloy. Aluminum bars, rods, and shapes shall be temper H14 or H24, conforming to QQ-A-200/8, QQ-A-225/1 or QQ-A-225/2. Aluminum alloy plate and sheet for the cassette back shall be temper H14 or H24, conforming with QQ-A-250/1 or QQ-A-250/2; for the cassette front, shall be temper H32 or H34 of QQ-A-250/8; or QQ-A-250/11, temper T6.

3.2.3 Plastic. Phenolic plastic material shall conform to the requirements of L-P-509, type I, grade XX, sheet. When tested as specified in 4.4.2.3, the plastic shall be free from foreign matter; the opacity to x-rays shall be uniform throughout. The plastic shall have a satin or a dull grain finish where the plastic surface is exposed on the finished cassette.

3.2.4 Lead foil. Lead foil shall conform to the material and hardness requirements of QQ-L-201, grade B. The lead shall be a minimum (min.) .005 inch thick. The thickness of lead shall not vary more than 10 percent within a particular cassette when tested as specified in 4.4.1.

3.2.5 Wool felt. Wool felt shall be longfibered, soft, woven felt. It shall be free from animal oil and any foreign matter which may cause deterioration of the intensifying screens or x-ray film.

3.2.6 Magnesium alloy. Magnesium alloy shall conform to the requirements of QQ-M-44 for chemical composition.

3.2.7 Hardware. Rivets used for fastening the front panel to the frame shall be oval head tubular rivets, nickel plated or chromium over nickel plated brass, or corrosion-resisting steel. Screws, "D" ring and its attachment bracket shall be corrosion-resisting steel.

3.3 Construction. The types I and II radiographic film cassettes shall consist of a hinged front, a back, and intensifying screens; the type II cassettes shall also be supplied with a grid. The cassettes shall be furnished with a film-lift of corrosion-resisting steel to automatically raise a corner of the film when the cassette is opened. The closed cassettes shall have no protrusions that interfere with their insertion and withdrawal from cassette changers, Bucky trays, and other cassette holding devices. The cassettes shall pass the drop test specified in 4.4.2.1.

3.3.1 Cassette front. The cassette front shall consist of a frame, and a front panel securely riveted to the frame.

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3.3.1.1 Frame. The frame shall be formed from corrosion-resisting steel channel mitered at the corners and welded on all butting/contacting corner surfaces. Aluminum stiffeners shall be secured to the inside of the frame by screws. Stiffeners shall fit snugly inside the frame, with not more than 1/64 inch space between butting parts. The frame of the cassette shall be flat within 1/32 inch when tested as specified in 4.4.2.2. The actual diagonal of the cassette shall not exceed the diagonal of a square cassette of maximum permitted length and width, nor shall it be less than the diagonal of a square cassette of minimum permitted length and width.

3.3.1.2 Front panel. The front panel shall be constructed of aluminum .064 inch thick (No. 14 Brown & Sharpe (B&S) gage); or magnesium, minimum 0.064 inch thick, No. 14 B&S gage, maximum 0.080 inch thick, No. 12 B&S gage. The magnesium panel may have a .031 ± .005 inch thick laminated phenolic type plastic facing. The outside of the aluminum or magnesium panel shall have a tough, adherent, durable coating. Front panels shall be flat within 1/32 inch when tested as specified in 4.4.2.2. When tested as specified in 4.4.2.3, the x-ray opacity of the front panel shall be uniform and free from signs of foreign matter. As viewed from the front of the cassette, the type I front panel shall be black, and the type II front panel shall be amber (see 6.6).

3.3.2 Cassette back. The cassette back (or door) shall consist of a metal back panel, and a sheet of lead foil mounted on the back panel under a sheet of wool felt, or neoprene rubber foam cushioning material. The back shall be hinged to the cassette front and shall be provided with spring locks and a D-ring to facilitate opening.

3.3.2.1 Cassette back panel. The cassette back panel shall be constructed of corrosion-resisting steel or aluminum. Corrosion-resisting steel shall be .0703 inch thick (No. 15, United States Standard (USS) gage). Aluminum shall be .0808 inch thick (No. 12, B&S gage). A 20 percent reduction in thickness is permissible if the panel is embossed to add strength.

3.3.2.2 Cassette liners. Wool felt, neoprene rubber foam, or polyurethane foam conforming to type I, Class 2 of MIL-P-26514, shall be securely and uniformly cemented to the inside back and cover of the cassette. The cushioning material shall provide a smooth, flat surface against which the intensifying screen is secured to enable uniform screen pressure against the radiographic film during use. The cushioning material shall be neatly installed and located such to prevent light leakage. The cassette shall be light tight when tested as specified in 4.4.2.4. Adhesives shall not contain any material which will have an injurious effect on the felt, foam, intensifying screens or x-ray film. The adhesive, felt, foam or lead foil shall leave no shadow or artifacts on radiographic film when tested as specified in 4.4.2.5.

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3.3.2.3 Hinges. Two corrosion-resisting steel floating type hinges shall be supplied. The hinges shall secure the cassette back to the frame. The hinges shall be located such to center the cassette back when closed. The hinge design shall allow the cassette back to open to a full 180°. The hinge shall allow enough self-adjustment so that intensifying screens can be installed. Hinges shall be located on the short side of the cassette.

3.3.2.4 Spring locks and film contact. Size 4 cassettes shall have three flat-strip spring crossbar locks made of corrosion-resisting steel. The locks shall be attached to the back by riveting and shall, when depressed and rotated, engage slots in the cassette frame to close and lock the cassette, and shall maintain evenly distributed pressure for uniform screen-film contact. Test for uniformity of film contact shall be conducted as specified in 4.4.2.6. The locks shall be designed to rotate 360° when the cassette back is open. Sizes 1, 2, and 3 cassettes shall have two spring locks. The locks shall exert a pressure from .1 to .3 pounds per square inch of film when tested as specified in 4.4.2.7. The locks shall show no signs of fatigue, breaking, permanent deformation or impairment of proper alignment and operation when tested as specified in 4.4.2.8.

3.3.2.5 D-ring. A corrosion-resisting steel D-ring shall be securely attached to the outside back by means of an attachment bracket. The bracket shall be mounted centrally on the end of the back opposite the hinges. The D-ring shall be provided with a leaf spring mounted under the attachment bracket to maintain the D-ring in any open position.

3.3.3 Intensifying screens. Each cassette shall be furnished with a pair of high speed intensifying screens mounted within the cassette. The screens shall conform to the requirements for type I of GG-S-176.

3.3.4 Grid for type II. Type II cassettes only, shall be furnished with a grid (see 6.5). The grid shall be the linear focused type having the following characteristics:

Ratio	5:1
Lines per inch	80
Focal range	28 to 72 inches
Thickness	.1 inch
Weight, per square inch	.14 ounce

The grid shall consist of narrow lead strips separated by fiber or plastic interspaces, transparent to x-rays. The entire grid shall be sealed in a strong, durable, low density aluminum envelope impervious to moisture. The grid shall be fitted snugly into the cassette frame, and shall be retained in position by the frame stiffeners. The grid shall be removable by removing the frame stiffeners. There shall be no fasteners in the grid area. The radiographs taken as specified in 4.4.1.3 shall be furnished with the grid from cassette.

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3.4 Dimensions. The inside and outside dimensions of the cassettes shall be as specified in ANSI PH 3.21, except that all cassettes delivered under a single contract shall have the same thickness ± .005 inch. The thickness shall be not more than .690 inch.

3.5 Weight. The weight of any cassette of the same type and size from one contract or purchase order shall be within 5 percent of the average of five randomly selected cassettes.

3.6 Identification marking. The outer surface of the cassette back shall be legibly and permanently marked with the contract or purchase order number, the name or registered trademark of the manufacturer, suitable serial number, the size of cassette in inches; and grid ratio, lines per inch and focal range, when applicable.

3.7 Finish. Corrosion-resisting steel, excepting hardware items, shall have a Commercial Standard No. 4 finish.

3.8 Welding. Welding shall be durable, sound in every detail and in accordance with latest practices recommended by the American Welding Society. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal and/or electrode where applicable to provide the best possible weld. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove those substances and leave the surface clean is satisfactory. All aluminum parts shall be welded within 48 hours after a chemical cleaning treatment. Parts must be chemically recleaned if the 48 hour time period is exceeded. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks which affect the quality, and have more strength than required for purpose intended.

3.9 Workmanship. The workmanship shall be first class throughout. The cassettes shall be of durable and rugged construction and free from defects which detract from the appearance or which may impair the serviceability. The cassette shall be free from sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for

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the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the cassette when a first article sample is required (see 3.1). This inspection shall include examination of 4.3 and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

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TABLE I. Sampling for examination

	Inspection level	AQL (defects per hundred units
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. Testing for two or more properties at the same time shall be acceptable. Acceptance number shall be zero regardless of lot size.

TABLE II. Sampling for tests

Component	Property	Requirement	Test method	Inspection level
Corrosion-resisting steel	Chemical properties	3.2.1	4.4.1.1	1/
Aluminum	Chemical and Physical Properties	3.2.2	QQ-A-250 QQ-A-225, as applicable	1/
Plastic	Chemical and Physical Properties	3.2.3	L-P-509	
	Freedom from foreign matter and uniformity of opacity	3.2.3	4.4.2.3	S-1
Lead foil	Material and hardness requirements	3.2.4	QQ-L-201	S-1
	Thickness	3.2.4	4.4.1.2	S-1, 2/
Magnesium Alloy	Chemical Composition	3.2.6	QQ-M-44	
Intensifying screens	All requirements	3.3.3	GG-S-176	
Grids	Ratio, lines per inch, Focal range	3.3.4	4.4.1.3	S-1
		3.3.4	4.4.1.3	S-1
End item	Drop test	3.3	4.4.2.1	S-1
	Flatness: of cassette frame	3.3.1.1	4.4.2.2	S-1
	of front panel	3.2.1.2	4.4.2.2	S-1
	X-ray opacity of front panel	3.3.1.2	4.4.2.3	S-1
	Light leakage	3.3.2.2	4.4.2.4	S-1
	Cassette liners	3.3.2.2	4.4.2.5	S-1
	Film contact	3.3.2.4	4.4.2.6	S-1
	Force to close spring locks	3.3.2.4	4.4.2.7	S-1
	Fatigue of spring locks	3.3.2.4	4.4.2.8	S-1

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1/ Four ounces of each lot shall be selected for chemical analysis.
2/ Unit of product for inspection purposes shall be square feet.

4.3 Examination. The cassettes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below.

TABLE III. Classification of defects

Categories and defects	
Major:	
101	Cassette not complete with all components.
102	Unit not free of burrs and sharp edges.
103	Cassette door, hinges, and spring locks do not operate as specified.
104	Front panel color not as specified.
105	Intensifying screens not correctly installed.
Minor:	
201	Cassettes do not rest evenly on a flat surface.
202	Unit not clean; not free of dirt, oil, and grease.
203	Any weld not continuous; not clean; not free of pitting, splatter and discoloration.
204	Finish not as specified.
205	Identification marking not present, properly located, legible, correct, or permanently affixed.
206	Rivets not properly applied.
207	Cassette liners not neatly installed.

4.3.2 Dimensional examination. The cassette shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 percent defective.

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TABLE IV. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

4.3.3.1 Examination of closure, waterproofing and banding of containers. When containers are required to comply with PPP-B-566, PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636, or PPP-B-676, examination for defects in closure, waterproofing and banding shall be in accordance with the appendix to the applicable box specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Unless otherwise specified herein, test radiographs shall be made at a focal spot to film distance of 40 + 1 inches, at 80KVP, with screen-type, medium speed x-ray film; and with an exposure time to produce a diffuse transmission density of 1.0 + 0.2 for 4.4.1.3, 4.4.2.3 and 4.4.2.5 and a density of 2 to 3 for 4.4.2.6. The x-ray tube shall be perpendicular to the center of the cassette. The nominal focal spot size of the x-ray tube shall be not more than 1.0 mm. A minimum of 2.3 mm of aluminum filtration shall be in the useful beam. The exposed x-ray film shall be developed in accordance with the manufacturer's instructions. The developed x-ray film shall be viewed with in unaided eye using a fluorescent type, x-ray film illuminator.

4.4.1 Materials.

4.4.1.1 Chemical composition. Tests for chemical analysis shall be conducted in accordance with methods 111.2 or 112.2 of Fed. Std. No. 151.

4.4.1.2 Thickness of lead. The thickness of the lead sheet shall be measured with a dial type indicating gage at a sufficient number of points to determine the minimum and maximum thickness values. Measurements shall be accurate within .0001 inch.

4.4.1.3 Grids. The grid ratio and lines per inch shall be determined by measurement from radiographs. Radiographs of each grid shall be made as specified in 4.4 at focal spot to film distances of 28 and 72 inches. The radiographs shall show that the space between lead strips is uniform in opacity and free from foreign matter. Radiographs of the grid shall be furnished with each type II cassette.

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4.4.2 End item.

4.4.2.1 Drop test. The selected sample shall be dropped on each corner from a height of 30 inches on a concrete floor. The cassette shall be dropped so that the plane of the cassette is perpendicular to the floor and the center of gravity of the cassette is directly over the corner which strikes the floor. For type II cassettes, the grid front shall be removed from the cassette prior to the drop test. Except for deformation at the impacted corner area, the cassette shall show no other sign of damage. After the drop test, the sample cassette shall be subjected to and pass the tests specified in 4.4.2.2, 4.4.2.4, and 4.4.2.6.

4.4.2.2 Flatness. The test for flatness shall be conducted on the open and closed cassette. The cassette frame shall rest on a flat test surface and any height variation shall be measured by passing feeler gages between the flat and the frame. The deviation from flatness of the front panel shall be determined by measuring the maximum distance between the front face and a straight edge placed at various angles on the front panel. The damaged corners resulting from the drop test shall not be placed on the flat test surface, and shall not be measured.

4.4.2.3 X-ray opacity of front panel. A non-screen type radiographic film shall be exposed in a cardboard, or plastic film holder, through the front panel as specified in 4.4. The developed film shall be examined for uniformity of opacity and for evidence of included foreign matter. Visible shadows or non-uniformity shall be cause for rejection.

4.4.2.4 Lightproof test. A high speed radiographic film, of the applicable size, shall be placed in the cassette, and all sides and edges of the cassette shall be exposed to a 100-watt incandescent lamp, operated at its rated voltage, at a distance of not more than 1 foot for not less than 1 minute. The developed test film shall be compared with a developed, unexposed film of the same lot. A density on the test film greater than that on unexposed film shall be cause for rejection. When viewed against the illuminator, the test film shall show no density variations.

4.4.2.5 Cassette liners. A radiographic film in the sample cassette shall be exposed as specified in 4.4. Shadows and artifacts on the test film shall be cause for rejection of the cassette.

4.4.2.6 Film contact. Contact shall be determined by using a flat sheet of non-magnetic, bare metal, .039 + .004 inch thick, continuously perforated with holes of .098 + 0, -.02 inch diameter, at intervals of .16 + 0, -.02 inch. A flat screen of 1/4 inch mesh, made up of not more than 0.032 inch diameter, non-magnetic, bare, unsoldered wire may be used as an alternate. To facilitate the measurement of density of the film exposed during the test, the test object may have a clear area, 1/2 to 1 inch

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diameter, near its center. The test object shall be placed on the cassette using a cone which covers the film area but equally cuts the four film corners. Exposure shall be made as specified in 4.4. Uniformity of film contact shall be indicated by a uniformly sharp exposure through the mesh. The test shall be repeated after exposure of the cassette to a minimum temperature of 120°F. and a relative humidity of 80 ± 2 percent for 48 hours. The intensifying screens may be removed during the conditioning period.

4.4.2.7 Force to close spring locks. The force in pounds, required to depress each side of each spring lock shall be determined. The total force to depress the spring locks shall be divided by the area of the back panel (in square inches) to obtain the pressure exerted on the film.

4.4.2.8 Fatigue of spring locks. Spring locks shall be flexed 500 times through an arc four times greater than that required to close the cassette. The angle shall be determined by measuring the angle of the open lock with the cassette back. Spring locks are to be tested before assembling to the cassette.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each cassette shall be wrapped in kraft paper or other suitable material and packaged in a box constructed in accordance with PPP-B-566, PPP-B-636, type CF, class domestic or PPP-B-676. Closure of the box shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Cassettes shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The number of cassettes specified in table V for the type and size shall be packed on edge, not laid flat in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. The number of cassettes specified in table V for the type and size shall be packed on edge not laid flat in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

TABLE V. Exterior container quantities

Type	Size	Quantity
II	2	12
II	4	6
I	1	12
I	2	12
I	3	6
I	4	6

5.2.3 Level C. Cassettes shall be packed on edge, not laid flat in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers to point of delivery called for the in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Each unit and exterior shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Each unit and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. Conventional cassettes (type I) covered by this specification are intended for use with x-ray systems equipped with a grid, or in techniques not requiring a grid. The grid front cassettes (type II) are suitable for grid techniques requiring moderate clean-up of stray radiation and wide positioning latitude with respect to x-ray tube alignment and focal spot to film distance. Type I and II cassettes have a thick lead foil in the cassette back which prevents their use with exit type phototimer pick-up tubes.

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6.2 Ordering data. Procurement of documents should specify the following:

a. Title, number and date of this specification.
b. Type and size required (see 1.2).
c. When a first article is required (see 3.1, 4.1.2.1 and 6.3).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Armed Services Procurement Regulations. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 This specification covers the following items shown in the Federal Supply Catalog:

NSN	Item Identification	Size	Type
6525-00-226-0918	CASSETTE, RADIOGRAPHIC FILM, Grid Front, 10 x 12 inches.	2	II
6525-00-226-0919	CASSETTE, RADIOGRAPHIC FILM, Grid Front, 14 x 17 inches.	4	II
6525-00-721-9813	CASSETTE, RADIOGRAPHIC FILM, CONVENTIONAL, 8 x 10 inches.	1	I
6525-00-721-9814	CASSETTE, RADIOGRAPHIC FILM, CONVENTIONAL, 10 x 12 inches.	2	I
6525-00-721-9815	CASSETTE, RADIOGRAPHIC FILM, CONVENTIONAL, 11 x 14 inches.	3	I
6525-00-721-9816	CASSETTE, RADIOGRAPHIC FILM, CONVENTIONAL, 14 x 17 inches.	4	I

6.5 An acceptable grid for type II cassette is the Liebel-Flarsheim Company's Microline, or equal.

6.6 The front panel of the cassette has an aluminum equivalency of approximately 1.6 mm. This and past specifications have attempted to maintain this property at a constant low value so that radiographic techniques would not have to be changed nor would the radiation dosage have to be increased.

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Army - MD
Navy - MS
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Preparing activity:
DLA-DM

Civil Agency Coordinating Activities:
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VA-DMS

Project No. 6525-1125

Orders for this publication are to be placed with General Services Administration, acting as agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 45 cents each.

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DOCUMENT IDENTIFIER AND TITLE

GG-C-131C CASSETTES, RADIOGRAPHIC FILM (MEDICAL)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3. IS THE DOCUMENT RESTRICTIVE?

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